

ECEN 4/5632 Problem Set 3 Solutions

from ClaudeAI

1. Ideal Sinc Reconstruction

Given: $x[n] = \delta[n] + 2\delta[n - 1]$, sample period T .

a) Reconstructed signal

Ideal bandlimited reconstruction replaces each sample with a shifted, scaled sinc:

$$x_r(t) = \sum_n x[n] \operatorname{sinc}\left(\frac{t - nT}{T}\right) = \operatorname{sinc}\left(\frac{t}{T}\right) + 2 \operatorname{sinc}\left(\frac{t - T}{T}\right)$$

b) Verify sample values

$$x_r(0) = \operatorname{sinc}(0) + 2 \operatorname{sinc}(-1) = 1 + 2(0) = 1\checkmark$$

(using $\operatorname{sinc}(-1) = \sin(-\pi)/(-\pi) = 0$)

$$x_r(T) = \operatorname{sinc}(1) + 2 \operatorname{sinc}(0) = 0 + 2(1) = 2\checkmark$$

This confirms the key sinc property: $\operatorname{sinc}(k) = 0$ for nonzero integers k , and $\operatorname{sinc}(0) = 1$, so the reconstruction passes exactly through each sample.

c) Value at $t = T/2$

$$x_r(T/2) = \operatorname{sinc}(0.5) + 2 \operatorname{sinc}(-0.5)$$

$$\operatorname{sinc}(0.5) = \frac{\sin(\pi/2)}{\pi/2} = \frac{2}{\pi}, \quad \operatorname{sinc}(-0.5) = \frac{\sin(-\pi/2)}{-\pi/2} = \frac{2}{\pi}$$

$$x_r(T/2) = \frac{2}{\pi} + 2 \left(\frac{2}{\pi}\right) = \frac{6}{\pi} \approx 1.9099$$

2. Zero-Order Hold Frequency Response

Given: $h_{\text{ZOH}}(t) = u(t) - u(t - T)$, a rectangular pulse of height 1 on $[0, T]$.

a) Direct integration

$$H_{\text{ZOH}}(j\Omega) = \int_0^T e^{-j\Omega t} dt = \left. \frac{e^{-j\Omega t}}{-j\Omega} \right|_0^T = \frac{1 - e^{-j\Omega T}}{j\Omega}$$

b) Separate into linear-phase and magnitude terms

Factor out the "half-delay" phase:

$$H_{ZOH}(j\Omega) = e^{-j\Omega T/2} \cdot \frac{e^{j\Omega T/2} - e^{-j\Omega T/2}}{j\Omega} = e^{-j\Omega T/2} \cdot \frac{2\sin(\Omega T/2)}{\Omega}$$

Writing the magnitude term as a normalized sinc:

$$H_{ZOH}(j\Omega) = T \operatorname{sinc}\left(\frac{\Omega T}{2\pi}\right) e^{-j\Omega T/2}$$

- **Linear-phase term:** $e^{-j\Omega T/2}$ -- phase = $-\Omega T/2$, linear in Ω , corresponding to a pure delay of $T/2$ (half a sample period), the group delay of the hold.
- **Magnitude term:** $T \operatorname{sinc}(\Omega T/2\pi)$ -- a sinc-shaped rolloff that suppresses high frequencies (the familiar "sinc droop" of a DAC).

c) Normalized magnitude in dB

Let $\omega = \Omega T$. Then $|H_{ZOH}|/T = |\operatorname{sinc}(\omega/2\pi)|$.

At $\omega = 0.5\pi$:

$$\frac{\omega}{2\pi} = 0.25, \quad \operatorname{sinc}(0.25) = \frac{\sin(0.25\pi)}{0.25\pi} = \frac{0.7071}{0.7854} = 0.9003$$
$$20\log_{10}(0.9003) \approx -0.91 \text{ dB}$$

At $\omega = \pi$:

$$\frac{\omega}{2\pi} = 0.5, \quad \operatorname{sinc}(0.5) = \frac{2}{\pi} = 0.6366$$
$$20\log_{10}(0.6366) \approx -3.92 \text{ dB}$$

(This is the well-known ZOH droop of about -3.9 dB at the Nyquist frequency.)

3. Uniform Quantization and SQNR

Range: $\approx -1 \text{ V}$ to $+1 \text{ V}$, full-scale span = 2 V .

a) Step size for $B = 8$

$$\Delta = \frac{2}{2^B} = \frac{2}{256} = \frac{1}{128} = 0.0078125 \text{ V}$$

b) Error variance and RMS error

$$\sigma_e^2 = \frac{\Delta^2}{12} = \frac{(0.0078125)^2}{12} = 5.086 \times 10^{-6} \text{ V}^2$$

$$\sigma_e = \sqrt{\sigma_e^2} = \frac{\Delta}{\sqrt{12}} = 0.002255 \text{ V } (\approx 2.26 \text{ mV})$$

c) SQNR for 8-bit full-scale sine

Standard result: for a full-scale sinusoid quantized with B bits (uniform-error model),

$$\text{SQNR}(\text{dB}) = 6.02 B + 1.76$$

$$\text{SQNR}_8 = 6.02(8) + 1.76 = 48.16 + 1.76 = 49.92 \text{ dB}$$

d) SQNR for 16-bit full-scale sine

$$\text{SQNR}_{16} = 6.02(16) + 1.76 = 96.32 + 1.76 = 98.08 \text{ dB}$$

e) Reconciling "96 dB" vs "98 dB" for 16-bit audio

These numbers are not contradictory because they answer **different questions**:

- "**~96 dB**" is the common rule-of-thumb figure $6.02 B \approx 96.3 \text{ dB}$. It represents the *dynamic range* of the quantizer in a loose sense -- essentially $20\log_{10}(2^B)$, the ratio of the full-scale range to the quantization step, without including the $+1.76 \text{ dB}$ correction factor. It's often quoted as a quick "6 dB per bit" estimate and is sometimes meant as a conservative, generic dynamic-range spec rather than a precise SQNR for one specific signal.
- "**~98 dB**" is the *exact* SQNR formula result for the specific case of a **full-scale sinusoidal** test signal, which includes the $+1.76 \text{ dB}$ term that comes from the ratio of a sine wave's mean-square power to its peak value ($V_{rms}^2 = V_{pk}^2/2$).

So the 96 dB figure is a rounded/approximate dynamic-range statement (or omits the sine-specific correction), while 98 dB is the precise theoretical SQNR for a full-scale sine tone. Both are legitimate -- they simply correspond to slightly different definitions/assumptions (generic dynamic range vs. exact full-scale-sinusoid SQNR), and real-world 16-bit audio numbers can also be affected by dither, noise shaping, and measurement weighting, which further explains why quoted "16-bit" figures vary in the literature.

4. Finite Sequence z-Transform

Given: $x[n] = 2\delta[n+1] - \delta[n] + 3\delta[n-2]$

z-Transform

$$X(z) = \sum_n x[n]z^{-n} = 2z^1 - 1 + 3z^{-2}$$

$$\boxed{X(z) = 2z - 1 + 3z^{-2}}$$

ROC

The sequence has both a positive-time term ($2z^1$, i.e. nonzero at $n = -1$) and negative-time-index terms extending to $n = 2$ ($3z^{-2}$). Because there is a term $2z$ that diverges as $|z| \rightarrow \infty$, and a term $3z^{-2}$ that diverges as $|z| \rightarrow 0$, the ROC is the entire z -plane except the origin and infinity:

$$\text{ROC: } 0 < |z| < \infty$$

Does the DTFT exist?

Since $x[n]$ has **finite support** (only 3 nonzero samples), the sum defining $X(e^{j\omega})$ is a finite sum and converges for every ω . The unit circle $|z| = 1$ lies within the ROC, so **yes, the DTFT exists** (it is simply $X(e^{j\omega}) = 2e^{j\omega} - 1 + 3e^{-j2\omega}$).

5. Right-Sided Exponential and Convergence

Given: $x[n] = (1.2)^n u[n]$

z-Transform via geometric series

$$X(z) = \sum_{n=0}^{\infty} (1.2)^n z^{-n} = \sum_{n=0}^{\infty} (1.2 z^{-1})^n$$

This geometric series converges when $|1.2 z^{-1}| < 1$, i.e. $|z| > 1.2$, giving:

$$X(z) = \frac{1}{1 - 1.2 z^{-1}}, \quad \text{ROC: } |z| > 1.2$$

Pole

Pole at $z = 1.2$

Does the DTFT exist?

No. The ROC $|z| > 1.2$ does **not** include the unit circle $|z| = 1$ (since $1.2 > 1$). Equivalently, $(1.2)^n u[n]$ grows without bound and is not absolutely (or even square) summable, so the DTFT does not converge.

6. Same Algebra, Different ROC

Given: $X(z) = \frac{1}{1 - 0.75 z^{-1}}$

a) ROC: $|z| > 0.75$

This ROC is the exterior of the pole, associated with a **right-sided (causal)** sequence:

$$x[n] = (0.75)^n u[n]$$

b) ROC: $|z| < 0.75$

This ROC is the interior of the pole, associated with a **left-sided (anticausal)** sequence:

$$x[n] = -(0.75)^n u[-n - 1]$$

Why the same rational expression gives two different answers

The algebraic expression $X(z)$ alone only fixes the poles/zeros; the **ROC** determines which side of each pole the power series is expanded on (causal vs. anticausal), so specifying the ROC is essential to uniquely invert a rational $X(z)$.

7. Partial Fractions with Two ROCs

$$\text{Given: } X(z) = \frac{1}{(1-0.5z^{-1})(1-0.8z^{-1})}$$

Partial fraction expansion

$$\text{Write } X(z) = \frac{A}{1-0.5z^{-1}} + \frac{B}{1-0.8z^{-1}}.$$

$$\text{Substituting } x = z^{-1}: 1 = A(1 - 0.8x) + B(1 - 0.5x)$$

- At $x = 1/0.5 = 2$: $1 = A(1 - 1.6) = -0.6A \Rightarrow A = -\frac{5}{3}$
- At $x = 1/0.8 = 1.25$: $1 = B(1 - 0.625) = 0.375B \Rightarrow B = \frac{8}{3}$

(Check: $A + B = -5/3 + 8/3 = 1$ “œ”, matching $X(z)|_{x=0} = 1$.)

$$X(z) = \frac{-5/3}{1 - 0.5z^{-1}} + \frac{8/3}{1 - 0.8z^{-1}}$$

Poles at $z = 0.5$ and $z = 0.8$.

a) ROC: $|z| > 0.8$ (**outside both poles -- causal/stable-looking, right-sided**)

Both terms are inverted as right-sided sequences:

$$x[n] = -\frac{5}{3}(0.5)^n u[n] + \frac{8}{3}(0.8)^n u[n]$$

b) ROC: $0.5 < |z| < 0.8$ (**between the two poles**)

This ROC lies **outside** the pole at $z = 0.5$ (so that term is right-sided) but **inside** the pole at $z = 0.8$ (so that term is left-sided/anticausal):

$$x[n] = -\frac{5}{3}(0.5)^n u[n] - \frac{8}{3}(0.8)^n u[-n - 1]$$

This is a genuinely **two-sided** sequence -- decaying for $n \rightarrow +\infty$ (from the 0.5-pole term) and also decaying for $n \rightarrow -\infty$ (from the 0.8-pole term), which is exactly what an annular ROC between two poles signals.